E016

Assignment 6

1. RP : $z(t) = x_1 \cos \omega_0 t - x_2 \sin \omega_0 t$

$\cancel{E[z(t)] = 0}$

$E[$

$E[X_1] = E[X_2] = 0$

$Var(X_1) = Var(X_2) = \sigma^2$

$$E[z(t)] = E[X_1 \cdot \cos \omega_0 t - X_2 \sin \omega_0 t]$$
$$= \cos(\omega_0 t) \cdot E[X_1] - \sin(\omega_0 t) \cdot E[X_2]$$
$$= \cos(\omega_0 t) \cdot (0) - \sin(\omega_0 t) \cdot (0)$$
$$= \boxed{0}$$

$$V[x_1 \cos \omega_0 t - x_2 \sin \omega_0 t] = $$
$$\cos(\omega_0 t)^2 \cdot Var(X_1) + (-\sin(\omega_0 t))^2 \cdot V[X_2]$$

$$= \sigma^2 (\cos^2(\omega_0 t) + \sin^2(\omega_0 t))$$
$$= \sigma^2 (1)$$
$$= \boxed{\sigma^2}$$

4. $X(t) = A \cdot \cos(\omega t)$

$A \sim U[0, 1]$

$$f(A) = \frac{1}{1-0} = 1$$

$$E[X(t)] = \int_0^1 A \cdot \cos(\omega t) \cdot 1 \, dA$$

$$= \cos(\omega t) \cdot \left[\frac{A^2}{2}\right]_0^1$$

$$= \cos(\omega t) \cdot \frac{1}{2}$$

ACR $\rightarrow R_x(t_1, t_2) = E[X(t_1) \cdot X(t_2)]$

$E[A^2] = E[(A \cdot \cos \omega t_1) \cdot (A \cos \omega t_2)]$

$= \cos(\omega t_1) \cdot \cos(\omega t_2) \cdot \int_0^1 A^2 \, dA$

$= \cos(\omega t_1) \cdot \cos(\omega t_2) \cdot \left[\frac{A^3}{3}\right]_0^1$

$= \boxed{\frac{1}{3} \cdot \cos(\omega t_1) \cdot \cos(\omega t_2)}$

ACV $\rightarrow C_x(t_1, t_2) = R_x(t_1, t_2) - E[X(t_1)] \cdot E[X(t_2)]$

$= \frac{1}{3} \cos(\omega t_1) \cdot \cos(\omega t_2) - \frac{1}{4} \cos(\omega t_1) \cdot \cos(\omega t_2)$

$= \frac{1}{3} - \frac{1}{4} \left(\cos(\omega t_1) \cdot \cos(\omega t_2)\right)$

$= \boxed{\frac{1}{12} \cos(\omega t_1) \cdot \cos(\omega t_2)}$

5. $X(t) = a \cdot \cos(bt + Y)$

$Y \sim U[-\pi, \pi]$

ACR $\rightarrow R_x(t_1, t_2) = ? = E[X(t_1) \cdot X(t_2)]$

$f(Y) = \frac{1}{\pi - (-\pi)} = \frac{1}{2\pi}$

$E[X(t)] = E[a \cdot \cos(bt + Y)]$

$= \frac{1}{2\pi} \int_{-\pi}^{\pi} a \cdot \cos(bt + Y) \, dY$

$= \frac{a}{2\pi} \left[\sin(bt + Y)\right]_{-\pi}^{\pi}$

$$= \frac{a}{2\pi}\left[\sin(bt+\pi) - \sin(bt-\pi)\right]$$

$$= \frac{a}{2\pi}\left[\cancel{\sin(bt)\cdot\cos(\pi)} + \cos(bt)\cdot\sin(\pi) - \cancel{\sin(bt)\cdot\cos(\pi)} + \cos(bt)\cdot\sin(\pi)\right]$$

$$= \frac{a}{2\pi}\left[2\cos(bt)\cdot\sin(\pi)\right] = 0$$

$$\therefore E[x(t)] = 0$$

$$ACR \rightarrow E\left[a\cdot\cos(bt_1+Y)\cdot a\cdot\cos(bt_2+Y)\right]$$

$$= a^2\cdot\frac{1}{2\pi}\int_{-\pi}^{\pi}\cos(bt_1+Y)\cdot\cos(bt_2+Y)\,dY$$

$$= \frac{a^2}{2\pi}\cdot\frac{1}{2}\int_{-\pi}^{\pi}\cos(bt_1+Y-bt_2-Y) + \cos(bt_1+Y+bt_2+Y)\,dY$$

$$= \frac{a^2}{4\pi}\int_{-\pi}^{\pi}\cos(bt_1-bt_2) + \cos(bt_1+bt_2+2Y)\,dY$$

$$= \frac{a^2}{4\pi}\left[\cos(bt_1-bt_2)\cdot Y + \frac{\sin(bt_1+bt_2+2Y)}{2}\right]_{-\pi}^{\pi}$$

$$= \frac{a^2}{4\pi}\left[\cos(bt_1-bt_2)\cdot\pi + \cos(bt_1-bt_2)\cdot\pi\right]$$

$$= \boxed{\frac{a^2}{2}\cos(bt_1-bt_2)}$$

6. $x(t) = \cos(\omega_0 t + \theta)$

$\theta \sim U[-\pi, \pi]$

First Order Moment $\rightarrow$ Mean $\rightarrow E[X(t)]$

$$E[X(t)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos(\omega_0 t + \theta)\, d\theta$$

$$= \frac{1}{2\pi}\left[\sin(\omega_0 t + \theta)\right]_{-\pi}^{\pi} = 0$$

Second Order Moment $\rightarrow E[X^2(t)]$

$$E[X^2(t)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos^2(\omega_0 t + \theta)\, d\theta$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\cos(2\omega t + 2\theta) + 1}{2}\, d\theta$$

$$= \frac{1}{4\pi}\left[\frac{\sin(2\omega t + 2\theta)}{2} + \theta\right]_{-\pi}^{\pi}$$

$$= \frac{1}{4\pi}\left[\frac{\sin(2\omega t + 2\pi)}{2} + \pi - \frac{\sin(2\omega t - 2\pi)}{2} + \pi\right]$$

$$= \frac{1}{2}$$

As $E[X(t)] = 0$ and $E[X^2(t)] = \frac{1}{2}$

$\therefore$ They are independent of time //

$$E(X') = E[\cos(\omega_0 t + \theta_0)] = \cos(\omega_0 t + \theta_0)$$

$\therefore$ Not independent of time.

7 $x(t) = a \cdot \cos(bt + Y)$

$Y \sim U[0, \pi/2]$

$$ACR = C_x(t_1, t_2) = R_x(t_1, t_2) - E[X(t_1)] \cdot E[X(t_2)]$$

$$E[X(t)] = \frac{2a}{\pi} \int_0^{\frac{\pi}{2}} \cos(bt + Y) \cdot dY$$

$$= \frac{2a}{\pi} \Big[\sin(bt + Y) \Big]_0^{\frac{\pi}{2}}$$

$$= \frac{2a}{\pi} \left[\sin\left(bt + \frac{\pi}{2}\right) - \sin(bt) \right]$$

$$= \frac{2a}{\pi} \left[\sin(bt) \cdot \cos\left(\frac{\pi}{2}\right) + \cos(bt) \cdot \sin\left(\frac{\pi}{2}\right) \right]$$

$$= \boxed{\frac{2a}{\pi} \cdot \cos(bt)}$$

$$R_x(t_1, t_2) = E(X(t_1) \cdot X(t_2))$$

$$= \frac{\cancel{2}a^2}{\pi} \int_0^{\frac{\pi}{2}} \cos(bt_1 + Y) \cdot \cos(bt_2 + Y)\, dY$$

$$= \frac{a^2}{\pi} \int_0^{\frac{\pi}{2}} \cos(bt_1 - bt_2) + \cos(2Y + bt_1 + bt_2)\, dY$$

$$= \frac{a^2}{\pi} \left[\cos(bt_1 - bt_2) Y + \frac{\sin(2Y + bt_1 + bt_2)}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{a^2}{\pi}\left[\cos(bt_1 - bt_2)\frac{\pi}{2} + \frac{\sin(\pi + bt_1 + bt_2)}{2} - \frac{\sin(bt_1 + bt_2)}{2}\right]$$

$\sin(\theta+\pi) = -\sin\theta$

$$= \frac{a^2}{\pi}\left[\cos(bt_1 - bt_2)\frac{\pi}{2}\right] - \frac{a^2}{\pi}\sin(bt_1 + bt_2)$$

$$\boxed{= \frac{a^2}{2}\cos(bt_1 - bt_2) - \frac{a^2}{\pi}\sin(bt_1 + bt_2)}$$

RVP 12/02

8. $x(t) = A\cos(2\pi t + Y)$

Y :	0	$\pi/2$
$P(Y=y)$:	$1/2$	$1/2$

$$E[x(t)] = \sum P(Y=y_i)\cdot x(t; y_i)$$

$$= \frac{1}{2}\cdot A\cos(2\pi t + 0) + \frac{1}{2}A\cos(2\pi t + \pi/2)$$

$$= \frac{1}{2}A\cos(2\pi t) - \frac{1}{2}A\sin(2\pi t)$$

$$\boxed{E[x(t)] = \frac{A}{2}[\cos(2\pi t) - \sin(2\pi t)]}$$

$$R_x(t_1, t_2) = E[x(t_1)\cdot x(t_2)]$$

$$= E[A\cos(2\pi t_1 + Y)\cdot A\cos(2\pi t_2 + Y)]$$

$$= A^2 E[\cos(2\pi t_1 + Y)\cdot\cos(2\pi t_2 + Y)]$$

$$= \frac{A^2}{2}E[\cos(2\pi t_1 + 2\pi t_2 + 2Y) + \cos(2\pi t_1 - 2\pi t_2)]$$

$$= \frac{A^2}{2}\cos(2\pi t_1 - 2\pi t_2) + \frac{A^2}{2}E[\cos(2\pi t_1 + 2\pi t_2 + 2Y)]$$

$$= \frac{A^2}{2}\cos(2\pi t_1 - 2\pi t_2) + \frac{A^2}{2}[\cos(2\pi t_1 + 2\pi t_2) + \cos(2\pi t_1 + 2\pi t_2 + \pi)]$$

$\cos 2x = 2\cos^2 x - 1$

$$= \frac{A^2}{2}\cos(2\pi t_1 - 2\pi t_2) + \frac{A^2}{2}\left[\cos(2\pi t_1 - \cancel{2\pi t_2}) - \cos(\cancel{2\pi t_1} - 2\pi t_2)\right]$$

$$= \boxed{\frac{A^2}{2}\cos(2\pi t_1 - 2\pi t_2)}$$

9. $x(t) = A\cdot\cos(\omega t + \theta)$

$\theta \sim U[-\pi, \pi]$

$R_{yy}(t_1, t_2) = ?$ $E[x(t_1)\cdot x(t_2)]$

$\rightarrow Y(t) = x^2(t)$

$$Y(t) = [A\cdot\cos(\omega t + \theta)]^2 = A^2\cdot\cos^2(\omega t + \theta)$$

$$E[Y(t_1)\cdot Y(t_2)] = E[A^2\cdot\cos^2(\omega t_1 + \theta)\cdot A^2\cdot\cos^2(\omega t_2 + \theta)]$$
$$= A^2 E[\cos^2(\omega t_1 + \theta)\cdot\cos^2(\omega t_2 + \theta)]$$
$$= \frac{A^2}{4} E[\cos(2\omega t_1 + 2\theta) + 1\cdot\cos(2\omega t_2 + 2\theta) + 1]$$

$$= \frac{A^2}{4}\cdot E[\cos(2\omega t_1 + 2\theta)\cdot\cos(2\omega t_2 + 2\theta) + \cos(2\omega t_1 + 2\theta) + \cos(2\omega t_2 + 2\theta) + 1]$$

$$= \frac{A^2}{4}\cdot E\left[\frac{1}{2}(\cos(2\omega t_1 + 2\omega t_2 + 4\theta) + \cos(2\omega t_1 - 2\omega t_2)) + \cos(2\omega t_1 + 2\theta) + \cos(2\omega t_2 + 2\theta) + 1\right]$$

$$= \frac{A^2}{4}\left[\frac{1}{2}E[\cos(2\omega t_1 + 2\omega t_2 + 4\theta) + \cos(2\omega t_1 - 2\omega t_2)] + E[\cos(2\omega t_1 + 2\theta) + \cos(2\omega t_2 + 2\theta) + 1]\right]$$

$$\cancel{= \frac{A^2}{4}\left\{\frac{1}{2\pi}\left[\sin\right.\right.}$$

$$= \frac{A^2}{8\pi}\left[\frac{1}{2}\left[\frac{\sin(2\omega t_1 + 2\omega t_2 + 4\theta)}{4} + \theta\cos(2\omega t_1 - 2\omega t_2)\right]_{-\pi}^{\pi} + \left[\frac{\sin(2\omega t_1 + 2\theta)}{2} + \frac{\sin(2\omega t_2 + 2\theta)}{2} + \theta\right]_{-\pi}^{\pi}\right.$$

$$= \frac{A^2}{8\pi}\left[\frac{1}{2}\left[\frac{\sin(2\omega t_1 + 2\omega t_2 + 4\pi)}{4} + \pi\cos(2\omega t_1 - 2\omega t_2) + \frac{\sin(4\pi - (2\omega t_1 + 2\omega t_2))}{4} + \pi\cos(2\omega t_1 - 2\omega t_2)\right]\right.$$

$$+\left[\frac{\sin(2\omega t_1 + 2\pi)}{2} + \frac{\sin(2\omega t_2 + 2\pi)}{2} + \pi + \frac{\sin(2\pi - 2\omega t_1)}{2} + \frac{\sin(2\pi - 2\omega t_2)}{2} + \pi\right]\Bigg]$$

$$= \frac{A^2}{8\pi}\left[\frac{1}{2}\left[2\pi\cos(2\omega t_1 - 2\omega t_2) + \cancel{\frac{\sin(2\omega t_1 + 2\omega t_2)}{4}} - \cancel{\frac{\sin(2\omega t_1 + 2\omega t_2)}{4}}\right.\right.$$

$$+\left[2\pi + \cancel{\frac{\sin(2\omega t_1)}{2}} + \cancel{\frac{\sin(2\omega t_2)}{2}} - \cancel{\frac{\sin(2\omega t_1)}{2}} - \cancel{\frac{\sin(2\omega t_2)}{2}}\right]\Bigg]$$

$$= \frac{A^2}{8\pi}\left[\pi\cos(2\omega t_1 - 2\omega t_2) + 2\pi\right]$$

$$= \frac{A^2}{8\cancel{\pi}}\left[\cancel{\pi}\left(2 + \cos(2\omega t_1 - 2\omega t_2)\right)\right]$$

$$= \boxed{\frac{A^2}{8}\left[2 + \cos(2\omega t_1 - 2\omega t_2)\right]}$$

10. $E[X(t)] = 3$

$R_X(t_1, t_2) = 16 + 4e^{-0.3(t_1 - t_2)}$

$Z = X(8) \rightarrow t = 8$

$E[X(8)] = \boxed{3} \rightarrow$ constant for all t

$R_X(8,8) = 20 = E[Z^2(8)]$

$V[X(8)] = E[Z^2(8)] - (E[Z])^2$

$= 20 - 9$

$= \boxed{11}$

11. $X(t) = A + Bt$

$E(A) = a, \quad E(B) = b$

$V(A) = \sigma^2_a, \quad V(B) = \sigma^2_b$

$E(X(t)) = E[A + Bt]$

$= E(A) + t \cdot E(B)$

$\boxed{E[X(t)] = a + tb}$

$R_x(t_1, t_2) = E[(A + Bt_1) \cdot (A + Bt_2)]$

$= E[A^2 + ABt_2 + ABt_1 + B^2 t_1 t_2]$

$= E[A^2] + t_2 \cdot E[A] \cdot E[B] + t_1 \cdot E[A] \cdot E[B] + t_1 t_2 \cdot E[B^2]$

$= t_2 \cdot a \cdot b + t_1 \cdot a \cdot b + E[A^2] + t_1 t_2 \cdot E[B^2]$

$= t_1 ab + t_2 ab + [V(A) + (E[A])^2] + t_1 t_2 \cdot [V(B) + (E[B])^2]$

$= t_1 \cdot ab + t_2 \cdot ab + \sigma^2 a + a^2 + t_1 t_2 \cdot \sigma^2 b + b^2$

$\boxed{R_x(t_1, t_2) = (\sigma^2 a + a^2) + ab(t_1 + t_2) + t_1 t_2 (\sigma^2 b + b^2)}$

$C_x(t_1, t_2) = R_x(t_1, t_2) - E[X(t_1)] \cdot E[X(t_2)]$

$E[X(t_1)] = a + t_1 b$

$E[X(t_2)] = a + t_2 b$

$E[X(t_1)] \cdot E[X(t_2)] = (a + t_1 b) \cdot (a + t_2 b)$

$= a^2 + abt_2 + abt_1 + ab^2 t_1 t_2$

$C_x(t_1, t_2) = \cancel{abt_1} + \cancel{abt_2} + \sigma^2 a + \cancel{a^2} + t_1 t_2 \sigma^2 b + \cancel{b^2 t_1 t_2}$
$- \cancel{a^2} - \cancel{abt_2} - \cancel{abt_1} - \cancel{b^2 t_1 t_2}$

$\boxed{C_x(t_1, t_2) = \sigma^2 a + \sigma^2 b \, t_1 t_2}$